

CASE STUDY

From Static Dashboard to Conversational Insight Tool

Adding a natural-language AI query layer to a Depression Risk Index Power BI dashboard

Role: Analyst & Product Builder

Tools: Power BI, Lovable (AI app builder), Claude API

Timeline: Self-directed

1. The Problem

The original capstone was a Power BI dashboard analyzing depression risk factors across a 413,768-row dataset. It lets users explore risk through slicers filtering by marital status, age, employment, and other variables one clicks at a time.

That interaction model has a ceiling, that is, it only answers the questions the slicers were built to anticipate. A user curious about a specific combination, “what’s the risk for widowed people over 60 with a chronic illness?” has to manually locate and combine the right filters, if that combination is even exposed at all. The dashboard rewards people who already know what to look for, and says little to people who don’t.

The goal is it allows users ask questions and get a direct answer without turning the dashboard into a black box that makes up numbers when it doesn’t actually know.

2. Approach

Rather than sending the full 413K-row dataset to an AI model on every question which would be slow, and prone to error, the dashboard’s own aggregation logic was reused to pre-compute a set of small summary tables. Each table crosses one or two variables (e.g., age group × chronic illness) against average risk score and sample size, so every number the AI can return is one that was already verified against the dashboard.

This grounded answers only constraint shaped every decision that followed:

- Every summary table keeps a count column, so any group with a small sample size can be flagged rather than trusted blindly.
- The AI is instructed to answer only from the uploaded tables, and to say so plainly when a question falls outside what the data covers rather than infer or estimate.
- Two-variable tables were used instead of wide, multi-way crosses, to avoid thin, statistically unreliable subgroups.

3. What Was Built

19 summary tables were derived from the original dataset, covering demographic (marital status, age), socioeconomic (employment, income, education), health (chronic illness, personal and family mental health history, substance abuse), lifestyle (diet, physical activity, sleep, smoking), and family (parenthood) factors.

These tables were uploaded into a Lovable-built web app alongside a text input and a Claude-powered backend. On each question, the app selects the relevant table(s), returns a short answer, the specific supporting statistic, and a

chart built from the matching rows reproducing what a user would otherwise have had to build themselves with the dashboard's slicers.

4. Key Findings Surfaced Through the Tool

Factor	Comparison	Risk Score Gap
Employment + Income	Employed/High Income vs. Unemployed/Low Income	5.42 → 9.37
Marital Status + Age	Widowed 60+ vs. overall average	8.43 (highest single group)
Chronic Illness	Married 60+, illness vs. no illness	7.58 → 8.59
Smoking	Current smoker, 45-59	8.63
Physical Activity	Sedentary, 18-29	8.44
Parenthood	With kids vs. without kids, low income	9.22 vs. 9.31 (no meaningful gap)

Two patterns stood out while building the summary tables. First, risk factors compound rather than act independently; unemployment and low income each raise risk on their own, but together (9.37) they exceed either alone. Second, not every assumption held: parenthood made almost no difference to risk once income was held constant, a useful check against over-attributing risk to any one variable without testing it directly.

5. Testing for Trustworthiness

Because the core design goal was a tool that doesn't fabricate an answer, testing focused less on "does it sound right" and more on two specific failure modes:

- Accuracy, that is, known values (e.g., avg. risk score for married, 60+, with chronic illness = 8.59) were checked against the tool's response and matched to three decimal places.
- Honest refusal, that is, questions outside the data's scope (e.g., "what's the risk for people who work night shifts?") or spanning a combination never crossed in any table (e.g., unemployed smokers) were confirmed to return a clear explanation of what the data does and doesn't cover, rather than a guessed number.

Both failure modes were tested repeatedly across different question phrasings, with consistent results.